

Multiple Choice (10 marks)

1. How can crosses involving strains of highly inbred mice be used to determine a genetic basis for transplant incompatibility?
 - (a) Transplant rejection is determined by the size difference between donor and recipient.
 - (b) Transplant acceptance is solely based on the health of the recipient.
 - (c) Transplant rejection (or acceptance) has a genetic basis.
 - (d) Transplant success is solely determined by environmental factors.
 - (e) Transplant outcomes can be predicted by the coat color of the mice.
2. Besides temperature, what other physical conditions must be taken into account for the growth of bacteria?
 - (a) Levels of oxygen, acidity or alkalinity of the medium
 - (b) Temperature changes, light exposure
 - (c) Atmospheric pressure, noise levels
 - (d) Nitrogen levels, radiation exposure
 - (e) Soil texture, light wavelength
3. Why is the organism that causes smallpox not used in the vaccine against smallpox?
 - (a) The organism is extinct in the wild and cannot be harvested for vaccine production
 - (b) The organism that causes smallpox is too costly to use in vaccine production
 - (c) The use of the actual virulent variola virus may be dangerous if too much is administered
 - (d) Smallpox virus is confined to research laboratories
 - (e) There is no specific treatment for smallpox
4. All of the following environmental conditions increase the rate of transpiration from plants EXCEPT
 - (a) high temperature
 - (b) high relative humidity
 - (c) low relative humidity
 - (d) wind
5. A plant that grows along a trellis exhibits

- (a) heliotropism
- (b) hydrotropism
- (c) thigmotropism
- (d) negative gravitotropism
- (e) chemotropism

True / False (10 marks)

Answer True or False

- 6. True or False: Shoot apical meristem formation occurs during embryogenesis, prior to seed formation.
- 7. True or False: Certain nerve gases can inhibit cholinesterase, leading to symptoms such as tremors, spasms, or even death in humans.
- 8. True or False: The speed of the action potential is the primary factor that determines the brain's ability to detect differences in stimulus intensity.
- 9. True or False: A deficiency in the pancreas can result in tetany due to its role in calcium regulation.
- 10. True or False: The absence of a Y chromosome is sufficient to determine maleness in mammals, as X chromosomes alone dictate gender.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the expected phenotypic ratios of progeny resulting from a cross between a pink-flowered snapdragon and a white-flowered snapdragon, providing a detailed analysis of the genetic mechanisms involved.
- 12. Discuss the concept of "ontogeny recapitulates phylogeny" in the context of vertebrate development from zygote to fetus, analyzing its implications for evolutionary biology.

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